DISTRICT-WISE ANALYSIS OF UJJWALA LPG CONNECTION UPTAKE IN WEST BENGAL

A Machine Learning Study of PMUY Connection Trends and Clean-Fuel Usage
2016–17 to 2022–23 (April–February)

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Course: B.Sc. (Hons.) Computer Science
Project type: Machine Learning Application Study

Academic Project Report

Abstract

This study examines district-wise uptake of Liquefied Petroleum Gas (LPG) connections released under the Pradhan Mantri Ujjwala Yojana (PMUY) in West Bengal. The analysis uses annual district-level connection-release data from 2016–17 to 2021–22, with an additional partial-year observation for 2022–23 (April–February), covering 23 districts. National Family Health Survey (NFHS-5) district fact-sheet estimates of households using clean fuel for cooking are included as a companion indicator where available; values are available for 20 districts. Linear and degree-two polynomial regression are fitted separately to district time series. K-Means and hierarchical clustering are used to group districts by their annual PMUY release profiles. Model comparison uses leave-one-out cross-validation (LOOCV) on the six complete financial years. The cross-validation results favour linear regression in 20 of 23 districts, with mean RMSE of 67,367.875 compared with 110,442.957 for polynomial regression. The mean cross-validation R^2 values are negative for both models, indicating weak out-of-sample predictive performance. K-Means and hierarchical clustering obtain silhouette scores of 0.432 and 0.407 respectively, with two clusters selected for K-Means. The findings describe differences in annual connection-release patterns and clean-fuel use, but do not establish that PMUY connections alone caused changes in household fuel use.

Keywords: PMUY; Ujjwala Yojana; LPG; West Bengal; district analysis; linear regression; polynomial regression; K-Means; hierarchical clustering; NFHS-5; clean cooking fuel.

1. Introduction

Access to clean cooking energy is important for household welfare, health and gender equality. Traditional solid fuels can expose household members, particularly those responsible for cooking, to indoor air pollution. The Pradhan Mantri Ujjwala Yojana (PMUY) was launched to expand access to LPG connections among eligible women from low-income households. However, the release of a connection does not by itself guarantee sustained LPG use, because refill affordability, access and household preferences also matter.

This project studies district-wise PMUY connection releases in West Bengal and compares district patterns using machine-learning methods. NFHS-5 clean-fuel estimates are used to provide a broader household-level context. The study is associated with Sustainable Development Goal (SDG) 7 (Affordable and Clean Energy), SDG 3 (Good Health and Well-being) and SDG 5 (Gender Equality).

Objectives

1. Examine annual district-wise PMUY connection releases.
2. Fit linear and polynomial trends for each district.
3. Group districts with similar annual release profiles using K-Means and hierarchical clustering.
4. Describe district variation in NFHS-5 clean-fuel usage.
5. Compare linear and polynomial models using cross-validation.

2. Related Work

The project is grounded in policy and survey sources rather than attempting to estimate the causal impact of PMUY. Government publications document the launch and implementation of PMUY, while NFHS district fact sheets provide household-level indicators including the use of clean fuel for cooking. Regression and clustering are standard exploratory methods for identifying temporal patterns and similarities across geographic units. In this study, these methods are applied descriptively to district-level annual connection-release counts.

3. Dataset and Preprocessing

3.1 Data sources

The primary dataset contains district-wise LPG connections released under PMUY for West Bengal. It includes seven financial-year fields: 2016–17, 2017–18, 2018–19, 2019–20, 2020–21, 2021–22 and 2022–23 (April–February). The dataset contains 23 districts. The companion NFHS-5 indicator is the percentage of households using clean fuel for cooking, extracted from district fact sheets. NFHS-5 values are available for 20 districts; Alipurduar, Jhargram and Kalimpong have missing values because the corresponding fact sheets were not available in the compiled dataset.

3.2 Data quality and preprocessing

The inspection found 23 unique districts and no duplicate district records. PMUY annual fields contain values for all 23 districts. NFHS-5 clean-fuel values are missing for three districts and are retained as missing rather than imputed or set to zero. Six complete financial years (2016–17 through 2021–22) are used for the principal regression and clustering time-series analysis. The 2022–23 observation is partial (April–February) and should be treated separately in descriptive reporting rather than as a full-year value. The 2020–21 zero values are retained as recorded in the supplied dataset; the interpretation depends on the source's reporting definition and should not be confused with missing data.

3.3 Dataset summary

Item	Description
Geographic coverage	23 districts of West Bengal
PMUY period	2016–17 to 2022–23 (April–February)
Complete years used for model fitting	2016–17 to 2021–22 (6 observations per district)
NFHS-5 clean-fuel coverage	20 districts; 3 missing
Duplicate district records	0
Main target for trend models	Annual PMUY connections released

4. Methodology

4.1 Linear regression

A separate simple linear regression model is fitted for each district, using financial year as the independent variable and annual PMUY connection releases as the dependent variable. The slope indicates the direction and rate of the fitted linear trend: a negative slope indicates a downward trend in annual releases, while a positive slope indicates an upward trend. RMSE measures the typical magnitude of prediction error, and R^2 describes the proportion of variation explained by the fitted model on the observations used for fitting.

4.2 Polynomial regression

A degree-two polynomial regression model is fitted for each district to allow a curved relationship between year and annual PMUY releases. Its training RMSE and R^2 are compared with the linear model. Because a polynomial model is more flexible, a lower training error alone is not evidence of better predictive performance; cross-validation is therefore used for the model comparison.

4.3 Clustering

K-Means clustering and agglomerative hierarchical clustering are applied to standardized district-wise annual PMUY release profiles. Standardization places yearly features on comparable scales before distance-based clustering. The K-Means cluster count is evaluated using silhouette scores for K = 2 to K = 5. The hierarchical dendrogram visualizes the nested similarity structure between districts. Cluster identifiers (0 and 1) are arbitrary labels, not rankings.

4.4 Evaluation metrics

RMSE is the square root of the mean squared prediction error. R^2 is calculated as 1 minus the ratio of residual sum of squares to total sum of squares; a negative value means the model predicts worse than a constant prediction equal to the evaluation-set mean. The silhouette score ranges from -1 to 1, with larger values generally indicating better separation and cohesion of clusters. Model comparison uses leave-one-out cross-validation (LOOCV): each of the six yearly observations is predicted using the other five.

5. Experimental Setup

The analysis was implemented in Python using pandas and NumPy for data handling, Matplotlib for plots, and scikit-learn/SciPy for regression, clustering and evaluation. The dataset was read from the combined Excel file. District-wise regression was performed using the six complete financial years. The 2022–23 April–February data was excluded from the six-point LOOCV comparison because it is not a complete financial year. The same six-year window was used to keep the model comparison consistent.

6. Results

6.1 Overall PMUY annual release pattern

The supplied statewide average trend graph shows the mean district-level annual releases rising from approximately 109,586 in 2016–17 to 130,675 in 2018–19, followed by a sharp decline to approximately 34,538 in 2019–20 and zero in 2020–21, then a rise to approximately 91,147 in 2021–22. The zero in 2020–21 is a recorded value in the dataset and should be verified against the original government source before making a policy explanation. The chart shown covers 2016–17 through 2021–22; the partial 2022–23 value is not shown.

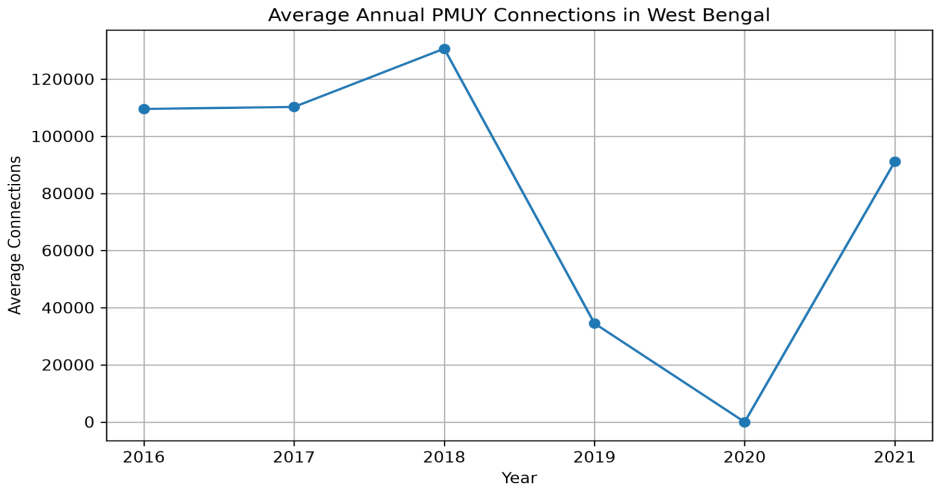


Fig. 1. Average annual PMUY connections across the 23 districts for the six complete financial years.

6.2 Linear and polynomial regression

The linear regression results show negative fitted slopes for most districts, indicating that annual connection releases generally declined over the fitted six-year period. The district-level RMSE and R^2 values vary considerably, reflecting differences in scale and temporal pattern. Polynomial regression reduces training error for some districts, but this does not guarantee better generalization.

Model	Mean training RMSE	Mean training R^2
Linear regression	42,829.56	0.299
Polynomial regression (degree 2)	39,947.10	0.384

Table 2. Average district-wise in-sample regression metrics. These are descriptive training-fit metrics and are not the cross-validation comparison.

6.3 Cross-validation model comparison

LOOCV favoured linear regression in 20 of 23 districts and polynomial regression in 3 districts. Mean LOOCV RMSE was 67,367.875 for linear regression and 110,442.957 for polynomial regression. Mean LOOCV R^2 was -0.702 and -3.538, respectively. Both average R^2 values are negative, indicating weak predictive performance on held-out annual observations. Given the very small sample of six years per district, these scores should be interpreted cautiously.

Metric	Linear	Polynomial
Mean LOOCV RMSE	67,367.875	110,442.957
Mean LOOCV R^2	-0.702	-3.538
Districts with lower LOOCV RMSE	20	3

Table 3. Leave-one-out cross-validation comparison using six complete financial years per district.

6.4 K-Means and hierarchical clustering

The silhouette analysis selected K = 2 as the best K-Means solution among the tested values. The K-Means silhouette score was 0.432. The hierarchical clustering silhouette score was 0.407. These values indicate some cluster structure, although separation is moderate rather than strong. The two methods do not produce identical memberships because their clustering procedures and linkage logic differ.

Clustering method	Selected clusters / setting	Silhouette score
K-Means	K = 2	0.432
Hierarchical	2-cluster cut for comparison	0.407

Table 4. Clustering quality scores.

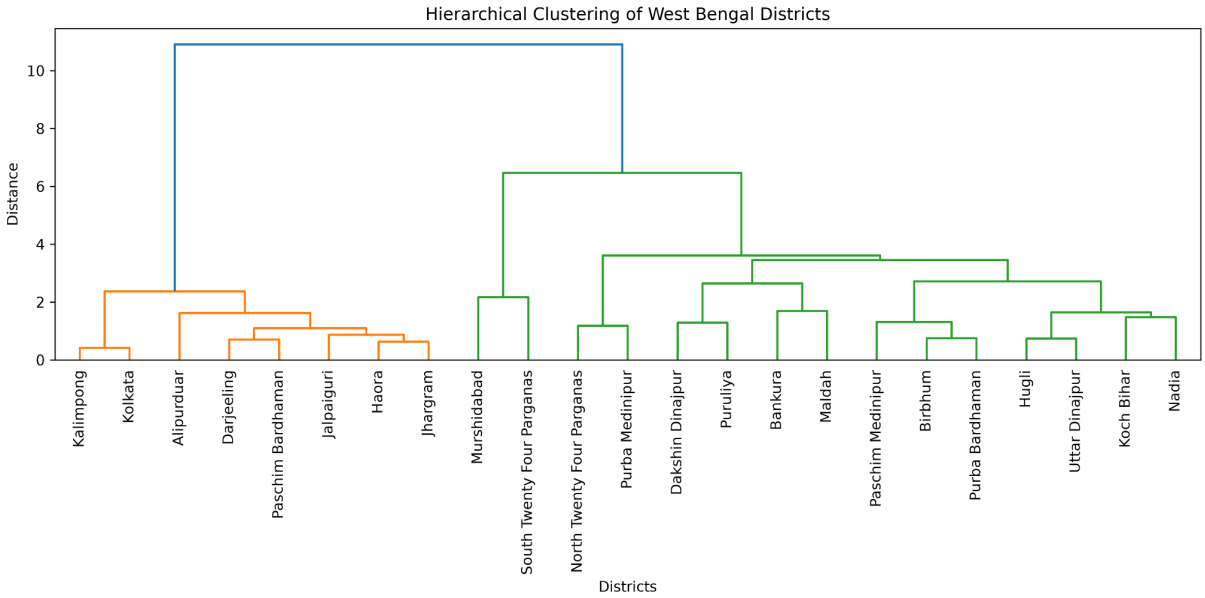


Fig. 2. Hierarchical clustering dendrogram for district-wise PMUY release profiles.

6.5 NFHS-5 clean-fuel usage

The NFHS-5 clean-fuel chart covers 20 districts with available values. The unweighted mean across these district percentages is 39.3%. Kolkata has the highest value in the supplied table (91.7%), while Purba Medinipur has the lowest (17.0%). This is a simple average of district percentages, not a population-weighted state estimate. Three districts are omitted because their NFHS-5 values are unavailable in the compiled dataset.

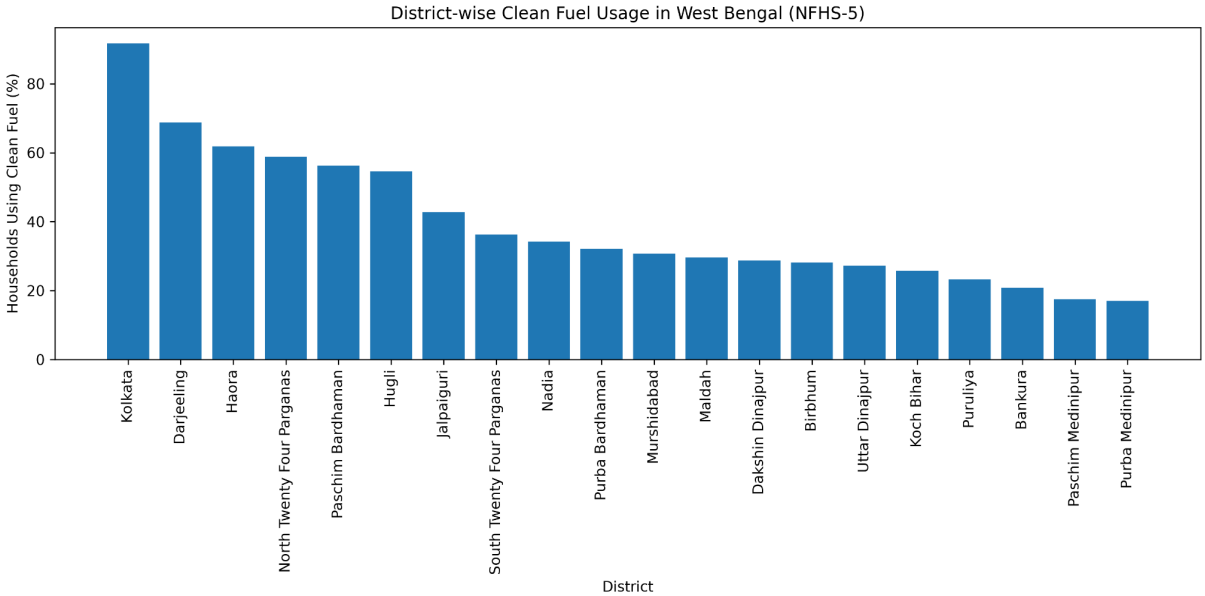


Fig. 3. District-wise percentage of households using clean fuel for cooking, NFHS-5.

7. Discussion

The analysis identifies substantial differences across districts in the scale and temporal pattern of PMUY connection releases. The regression slopes provide a compact description of district trends, while clustering groups districts with similar standardized release profiles. The NFHS-5 indicator adds a household-level view of clean-fuel use and shows that clean-fuel usage varies markedly across districts. However, PMUY connection counts and NFHS clean-fuel percentages measure different things: the former counts connections released under a scheme, while the latter estimates household use of clean cooking fuel. The two should not be treated as interchangeable or as direct evidence of causal impact.

The cross-validation comparison favours the simpler linear model on average, but both models have negative mean held-out R^2. This is not a software error: with only six annual observations per district and sharp changes in the series, predicting a held-out year is difficult. The negative scores indicate that the models are not reliable forecasting tools with this limited time series. They remain useful here as exploratory trend summaries, provided this limitation is clearly stated.

8. Limitations

1. The time series is short, with only six complete financial-year observations per district for regression validation.
2. The 2022–23 data cover April–February and are not a full financial year.
3. The 2020–21 zero values require confirmation from the original source and should not be interpreted as missing observations.
4. NFHS-5 clean-fuel data are unavailable for three districts in the compiled dataset.
5. The district-level NFHS percentage is a survey estimate and is not directly comparable to annual PMUY connection-release counts.
6. The clustering results depend on feature selection, scaling, distance metric and cluster cut; silhouette scores indicate moderate separation.
7. The analysis is descriptive and does not establish a causal relationship between PMUY and clean-fuel use.

9. Conclusion and Future Work

This project analysed PMUY LPG connection releases across 23 districts of West Bengal and examined NFHS-5 clean-fuel usage for 20 districts. Linear and polynomial regression described district-level trends, while K-Means and hierarchical clustering identified groups of districts with similar annual release profiles. In LOOCV, linear regression had lower RMSE in 20 districts, but negative average R^2 values for both models show that predictive performance is weak with the available six-year series. The K-Means solution with two clusters achieved a silhouette score of 0.432, slightly above the hierarchical score of 0.407. NFHS-5 clean-fuel percentages showed considerable variation across the districts with available data.

Future work could incorporate longer time series, population or eligible-household denominators, LPG refill/consumption indicators, rural–urban splits and district-level socioeconomic variables. A fuller analysis could also examine the relationship between connection release and sustained LPG use, while accounting for survey uncertainty and district population weights.

10. Declarations

Data availability: PMUY district-wise LPG connection data are available through the Open Government Data (OGD) Platform, India, under the Ministry of Petroleum and Natural Gas. NFHS-5 district fact sheets are published by the International Institute for Population Sciences (IIPS) and the Ministry of Health and Family Welfare.
Conflict of interest: The author declares no conflict of interest.
Funding: No external funding was received.
Code availability: Python scripts and generated result files are maintained in the project GitHub repository.

11. References

[1] Ministry of Petroleum and Natural Gas, Government of India, “District-wise details of Liquefied Petroleum Gas (LPG) connections released under Pradhan Mantri Ujjwala Yojana,” Open Government Data Platform India. Available: https://ap.data.gov.in/resource/district-wise-details-liquefied-petroleum-gas-lpg-connections-released-under-pradhan

[2] Ministry of Petroleum and Natural Gas, Government of India, Rajya Sabha, Unstarred Question No. 2217, answered 20 March 2023, on Pradhan Mantri Ujjwala Yojana.

[3] International Institute for Population Sciences (IIPS) and Ministry of Health and Family Welfare, Government of India, National Family Health Survey (NFHS-5), 2019–20: District Fact Sheets, West Bengal.

[4] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

Appendix A. Project files and reproducibility

The GitHub project contains the Python scripts used for data inspection, preprocessing, linear regression, polynomial regression, clustering, cluster analysis, NFHS-5 analysis and model comparison. The results folder contains CSV/Excel outputs and the generated figures. The execution logs preserve the terminal outputs. The complete source code is available in the linked repository; it is not reproduced in full here to keep the journal concise.

Note: Before final submission, verify the original source's explanation for the 2020–21 zero values and ensure the GitHub repository URL is inserted in the Code availability statement.